

Woojin Bae

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RESEARCH INTERESTS

Design and Discovery of Nanomaterials through Advanced Techniques

- **Nanomaterials:** Perovskite, Zeolite, Metal Oxide, Nanoparticle.
- **Advanced Techniques:** Transmission Electron Microscopy, Self-Driving Labs, DFT.

EDUCATION

Seoul National University (SNU), Seoul, South Korea

Expected Aug. 2026

B.S. Chemical and Biological Engineering (CBE), *Cum Laude (expected)*

- GPA: 3.72/4.3 (overall), 3.67/4.0 (converted)
- Graduate Course: Topics in Inorganic Material and Semiconductor Processes (Fall 2025)

North Carolina State University (NCSU), Raleigh, NC

May 2025

Exchange student in Chemical and Biomolecular Engineering

- GPA: 4.0/4.0, AIChE at NC State

Daejeon Science High School for the Gifted, Daejeon, South Korea

Feb. 2020

PUBLICATIONS & PRESENTATIONS

(†: equally contributed, *: corresponding author)

[P1] Jinge Xu, Junbin Li, **Woojin Bae**, Milad Abolhasani. *, Autonomous Mn-doped Perovskite Nanocrystals synthesis by a Self-Driving lab. *Manuscript in preparation.*

[P2] **W. Bae**†, S.Jung†, S. Ji†, Y. Jeon* & J. Park*, SEI Structures Dependent Li-Morphology Characterized by cryo-TEM. [Poster Presentation]

- 2025 KICChE Fall meeting and International Symposium, Oct. 15-18, 2025.
- The 2024 Annual Fall Conference of Korean Society of Microscopy, Nov. 25-26, 2024.

RESEARCH EXPERIENCE

Multi-dimensional Materials Chemistry Lab, SNU

Mar. 2024 – Present

Undergraduate Researcher - (PI: Prof. Jungwon Park)

Seoul, Korea

Topic: SEI Structures Dependent Li-Morphology Characterized by cryo-TEM

- Team Leader; Developed a DBSCAN-based protocol for quantitative image analysis for cryogenic TEM characterization of SEI structures. [P2]
- Generalized correlation between SEI morphology, crystallinity, and composition; proposed a formation mechanism of Li-deposition.

Topic: Computational Modeling of AI-Driven Synthesized Perovskite Nanocrystals

- Awarded a \$2,000 research grant by SNU College for a project across SNU and NCSU.
- Initiated and led collaborative project characterize to AI-optimized synthesized Mn-doped CsPb(Cl/Br)₃ QDs, by TEM analysis and DFT calculation.

Self-Driving Fluidic Lab, NCSU

Jan. 2025 – May 2025

Undergraduate Researcher - (PI: Prof. Milad Abolhasani)

Raleigh, NC

Topic: Autonomous Mn-doped Perovskite Nanocrystals Synthesis by a Self-Driving Lab

- Achieved Photoluminescence Quantum Yield improvement of Mn-doped CsPb(Cl/Br)₃ perovskite through genetic algorithm-based synthesis optimization in autonomous lab.
- Presented a poster at Spring Undergraduate Research & Creativity Symposium. [P1]

Summer Undergraduate Initiative (SURI), ASU
Undergraduate Researcher - (PI: *Prof. Sandhya Susarla*)

May 2025 – Jul. 2025
Tempe, AZ

Topic: Synthesis of Perovskite QDs for Phonon-Polariton Interaction with hBN

- Selected for the SURI fellowship at Arizona State University (\$5,000).
- Synthesized perovskite nanocrystals with controlled morphologies to investigate phonon–polariton coupling with hexagonal boron nitride (hBN).
- Presented a poster at Summer Research Initiative Final Research Showcase.

HONORS & AWARDS

Honorable Mention | Korea Process Simulation Olympiad 2025, KIChE, Korea *Sep. 2025*
• National finalist with process design for high-purity methanol (PI: *Prof. Jong Min Lee*)

Excellence Research Award | 2024 Student-Directed Research Symposium *Jan. 2025*
• Selected as a Top 3 project (out of 20) for an independent research study. [P2]

KorUS Educational Exchange Initiative for Youth in STEM | MOTIE, Korea *Nov. 2024*
• Awarded national \$9,000 scholarship as a STEM-specialized exchange student.

Excellence Award | Undergraduate Research Program for Advanced Equipment *Jan. 2024*
• Analyzed microscopic–macroscopic structure correlation in hollow ZSM-5 synthesis.

Future Chemical Talent Award | Korean Union of Chemical Science & Technology *Feb. 2020*
• Awarded as one of the top 6 science high school students in Korea.

LEADERSHIP & MEMBERSHIPS

Undergraduate Member | Korean-American Scientists and Engineers Association *Jan. 2025*
• Delivered a research presentation at Korean-American undergraduate conference.

Tokyo Forum Youth Session | The University of Tokyo, Japan *Fall 2024*
• Youth Representative of Korea, in a session on Aging Society and Low Birth Rate.

Young Engineers Honor Society | National Academy of Engineering of Korea *Nov. 2024*
• Korean engineering honor society under National Academy of Engineering of Korea.

Republic of Korea Air Force | 11th Fighter Wing, Daegu, Korea *Aug. 2022 – May 2024*

Marketing team manager | SNUfestival, Student Council, SNU *Jun. 2021 – May 2022*
• Promoted the festival through the online website and secured \$2,200 sponsorship from a shared scooter company during the first in-person festival after the COVID-19 pandemic.

Vice president | Chemical and Biological Engineering Interim Leadership, SNU *Winter 2020*
• Developed an online program for incoming freshmen during the COVID-19 pandemic.

TEACHING & MENTORING EXPERIENCES

SNU Corps in Indonesia | Universitas Mataram, Lombok, Indonesia *Feb. 2026 (Scheduled)*

Korean Language Peer Mentor | Language Education Institute, SNU *Fall 2024*

Peer Tutor | Basic Computing: First Adventures in Computing, SNU *Fall 2024, 2025*

Student Mentor, SNU Mentoring | SNU Social Responsibility *Apr. 2023 – Aug. 2025*

TECHNICAL SKILLS & LANGUAGES

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- **Instruments:** TEM, SEM, XRD, STEM/EDS, UV-Vis, OM, PL
 - **Programming Language:** Python, MATLAB, C
 - **Software:** Gatan DigitalMicrograph, VASP, APS, Adobe, Fusion 360
 - **Languages:** Korean (Native), English (C1)